

Saturated Image Escape and GCD Trace Clusters in a Growing Open Walsh Gate

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Abstract

For a three-symbol open Walsh factor shift, we determine the image rank at every system size and time. At time proportional to tensor length, the signed log-survival rate is linear and then saturates after one full factor cycle. Fixed-period traces retain a greatest-common-divisor dependence on the system size and need not converge, but dimension-normalized traces vanish. These are source-side large-system statements, not a secular, self-adjoint, or target-matched limit.

1 Frozen gate and exact power rank

Let $\omega = e^{2\pi i/3}$,

$$(F_3)_{j\ell} = 3^{-1/2}\omega^{j\ell}, \quad P = \text{diag}(1, 0, 1), \quad A = F_3^* P.$$

On $\mathcal{H}_k = (\mathbb{C}^3)^{\otimes k}$ freeze

$$B_k(v_0 \otimes \cdots \otimes v_{k-1}) = v_1 \otimes \cdots \otimes v_{k-1} \otimes Av_0. \quad (1)$$

One application is one clock tick. Direct calculation gives

$$\det(\lambda I - A) = \lambda(\lambda^2 - \tau\lambda + q_0), \quad \tau = \frac{\sqrt{3}}{6} - \frac{i}{2}, \quad q_0 = -\frac{1}{2} - \frac{\sqrt{3}i}{6}.$$

Since $q_0 \neq 0$, zero is a simple eigenvalue and the other two eigenvalues are nonzero. Hence $\text{rank}(A^m) = 2$ for every $m \geq 1$, while $\text{rank}(A^0) = 3$.

Theorem 1. *For every $k \geq 1$ and $n \geq 0$,*

$$\text{rank}(B_k^n) = 2^{\min(n,k)} 3^{k-\min(n,k)}. \quad (2)$$

Proof. Write $n = qk + r$, with $0 \leq r < k$. Iterating (1) on a pure tensor gives

$$B_k^n(v_0, \dots, v_{k-1}) = (A^q v_r, \dots, A^q v_{k-1}, A^{q+1} v_0, \dots, A^{q+1} v_{r-1}).$$

If $q = 0$, exactly $n = r$ factors have positive powers of A ; if $q \geq 1$, all k do. Tensor-rank multiplicativity proves (2), including the identity boundary $n = 0$. $\square$

The survival fraction is therefore

$$S_k(n) = \frac{\text{rank}(B_k^n)}{3^k} = \left(\frac{2}{3}\right)^{\min(n,k)}. \quad (3)$$

For $n_k = \lfloor \alpha k \rfloor$ and $\alpha \geq 0$,

$$\lim_{k \rightarrow \infty} \frac{1}{k} \log S_k(n_k) = \min(\alpha, 1) \log(2/3). \quad (4)$$

Thus the positive escape exponent is $E(\alpha) = \min(\alpha, 1) \log(3/2)$. At $\alpha = 0$ one has $n_k = 0$, $S_k = 1$, and $E(0) = 0$; no positive-time formula is silently substituted at this identity boundary.

2 Fixed-period traces

Let $t_0 = 2$, $t_1 = \tau$, and $t_m = \tau t_{m-1} - q_0 t_{m-2}$. Basis-index contraction along the cycles of addition by n modulo k gives, for $d = \gcd(n, k)$,

$$\text{Tr}(B_k^n) = t_{n/d}^d. \quad (5)$$

Indeed the factor permutation has d independent cycles, and each cycle contracts n/d ordered copies of A .

At fixed n , (5) ranges over the equality-merged set of divisor values

$$\{t_{n/d}^d : d \mid n\}. \quad (6)$$

where divisors that produce the same complex number define one cluster value. For each divisor $d \mid n$, the unbounded sequence $k_j = d(1 + j(n/d))$ satisfies $\gcd(n, k_j) = d$, so every divisor class occurs infinitely often. Conversely no other value can occur. Because this is a finite set,

$$3^{-k} \text{Tr}(B_k^n) \longrightarrow 0 \quad (7)$$

for every fixed period.

The normalization cannot be removed in general. At period two, odd k have $d = 1$ and even k have $d = 2$, so

$$t_2 = \frac{5}{6} + \frac{\sqrt{3}}{6}i, \quad t_1^2 = -\frac{1}{6} - \frac{\sqrt{3}}{6}i, \quad (1)$$

$$t_2 - t_1^2 = 1 + \frac{\sqrt{3}}{3}i = -2q_0 \neq 0. \quad (8)$$

The odd and even subsequences are distinct constants. Hence $\text{Tr}(B_k^2)$ has no unnormalized $k \rightarrow \infty$ limit.

3 Validation and boundary

The exact payload contains 624 rank rows, 192 macroscopic rows, and divisor ledgers through period twenty. A producer-independent checker, separate symbolic reconstruction, byte replay, and semantic mutation suite validate the implementation. The closed $P = I_3$ parent is unitary and has zero image escape.

The strict Route-A tuple is

$$(A1_WEAK, A2_FAIL, A3_FAIL, A4_UNITARY_OR_SCATTERING_CANDIDATE),$$

overall ROUTE_A_EXPLORATORY. We claim no full growing- k secular limit, self-adjoint quantization, target divisor, functional equation, counting law, prime-like map, arithmetic local data, Euler factor, root number, automorphy, Hilbert–Pólya operator, or Route-B authorization. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.